DATA ANALYSIS REPORT

Decision Process Engineering Capstone

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MULTIPLE LINEAR REGRESSION ANALYSIS

Quantifying Intervention Methods' Impact on Student Math Scores Improvement

Contents

A. RESEARCH QUESTION

The scope of this research paper is to examine whether a multiple linear regression model can be constructed from an education intervention dataset to statistically quantify the impact of different academic intervention methods on student math score improvement. This paper first describes the data collection and preparation methods, then presents the analytical results, discusses their implications for education, and concludes with recommendations for future research. With a dataset of 10,000 student records containing measurable academic indicators, the conditions necessary for regression analysis were sufficiently met

This study exists within the field of educational data mining and data analytics, where statistical modeling is increasingly used to evaluate academic interventions and predict student outcomes. Many self-management, peer-mediated, and adult-mediated interventions exist to promote academic achievement for youth across the age and ability spectrum. Self-monitoring activities, peer tutoring, and home-school collaborations are just a few examples of academic interventions that can be implemented in public schools to reach students who are academically at risk (Arrington, 2024).

The null hypothesis is that a multiple regression model constructed on the education intervention dataset would not statistically achieve a predictive accuracy exceeding 70% (R-Squared ≤ 0.70) in predicting math score improvement. The alternate hypothesis is that the model statistically will achieve a predictive accuracy exceeding 70% (R-Squared > 0.70). The 70% threshold was selected as a meaningful benchmark for predictive accuracy as it represents a model that explains a substantial portion of the variance in student math score improvement.

The research is justified by the increasing availability of student performance data and the widespread implementation of varied academic intervention methods across schools. Multiple linear regression evaluates the relationship between multiple independent variables and a single dependent variable, producing interpretable coefficients that indicate the magnitude of each predictor's contribution (Aissaoui et al., 2020).

B. DATA COLLECTION PROCESS

The data for this study were collected from Kaggle *(Kaggle, 2026)*, a publicly available platform widely used in academic research and data science. The education intervention dataset (Dr. Sewell W, 2026) is a publicly available synthetic educational dataset comprising 10,000 student records designed to simulate real-world academic performance data across multiple schools and grade levels. The dataset contains 16 variables capturing academic, behavioral, and instructional characteristics. The variables relevant to this study included *intervention type,* *attendance rate, weekly study hours, teacher experience years, class size, baseline math scores, socioeconomic index, baseline motivation,* and *test anxiety*. The dependent variable, *math_improvement*, was engineered by calculating the difference between each student's post-intervention math score and their baseline math score, directly measuring the academic gain attributable to each intervention method. Four intervention types were represented in the dataset: *blended, tutoring, flipped*, and a *control* group receiving traditional instruction. The variables captured are listed in the data table below.

DATA TABLE

COLUMN NAME	TYPE	SAMPLE DATA
student_id	Continuous	93207890110
school_id	Continuous	110
grade_level	Continuous	9
intervention	Categorical	blended
ses_index	Continuous	1.379
attendance_rate	Continuous	0.996
study_hours_wk	Continuous	5.93
teacher_experience_years	Continuous	6.7
class_size	Continuous	32
baseline_math	Continuous	78.9
baseline_reading	Continuous	82.3
baseline_motivation_1_10	Continuous	7.1
post_math	Continuous	88.9
post_reading	Continuous	95.2
test_anxiety_1_10	Continuous	3.8
passed	Continuous	1

An advantage of using a secondary data source such as Kaggle is the immediate availability of a large, structured dataset. With 10,000 student records already compiled and formatted, the time and resources typically required for primary data collection, such as surveys, institutional approvals, and participant recruitment, were eliminated entirely. This allowed the study to focus exclusively on data preparation, analysis, and interpretation, making the research process more efficient and reproducible. Additionally, the dataset's public availability ensures that the findings of this study can be independently verified and replicated by other researchers.

A disadvantage of using secondary data is the lack of control over how the data was originally collected. The dataset does not provide information on how students were assigned to intervention groups, raising concerns about selection bias. If students were not randomly assigned, pre-existing differences between groups, such as prior academic achievement or socioeconomic background, may have influenced the outcomes independently of the intervention method. Furthermore, the dataset excludes qualitative factors such as parental involvement, student mental health, and home environment, which are well-documented influences on academic performance and likely contribute to the low R-Squared of 13.75% observed in the final model.

As this study used a publicly available secondary dataset containing no personally identifiable information, no ethical approval was required. All data were handled in accordance with standard data privacy practices, and no individual student records were shared or published.

C. DATA EXTRACTION AND PREPARATION

The dataset was downloaded from Kaggle *(www.kaggle.com)* in comma-separated format and imported directly into RStudio. RStudio was used as the primary tool for all stages of data extraction and preparation because it is specifically designed for statistical computing and data analysis. It supports reproducible research through R Markdown and is open source, widely adopted in academic research and data science. All stages of analysis, including data cleaning, transformation, visualization, and reporting, can be performed within a single environment. Packages used during data preparation included *tidyverse* for data manipulation, *ggplot2* for data visualization and exploratory analysis, *car* for VIF-based multicollinearity diagnostics, and *caret* for model evaluation, cross-validation, and R-Squared assessment (Wickham, H. 2016).

The dataset was loaded into RStudio using the *read.csv ()* function.

```
### Import the 'CSV' file
```{r import_data}
student <- read.csv("education_intervention_dataset.csv")
```
```

Prior to any transformation, the dataset was inspected to verify data integrity using the *sum(is.na()), sum(duplicated()),* and *dim()* functions. This confirmed that the dataset contained 10,000 rows and 16 columns, with no missing or null values and zero duplicate records. The data sparsity rate was 0.84%, indicating that the dataset was sufficiently complete for analysis and that no imputation was necessary (Wickham, H. 2016).

```
### Check for missing or duplicate values in dataset
```{r}
sum(is.na(student))
sum(duplicated(student))
dim(student)
```

 [1] 0
 [1] 0
 [1] 10000   16
```

A new dependent variable, *math_improvement*, was engineered by calculating the difference between each student's post-intervention math score and their baseline math score. This variable directly measures the academic gain attributable to each intervention method, making it the most appropriate dependent variable for this study.

```
## DATA CLEANING
### Create 'math_improvement' variable
```{r math_improvement}
student$math_improvement <- student$post_math - student$baseline_math
```
```

The intervention variable contained four categorical text values: *blended, tutoring, flipped,* and *control*. Since multiple linear regression requires numerical inputs, label encoding was applied using the as.numeric(as.factor()) function to convert the categories to numerical values: blended = 1, control = 2, flipped = 3, and tutoring = 4.

```
### Label encode 'intervention' variable
```{r label_encode}
student$intervention <- as.numeric(as.factor(student$intervention))
```
```

Seven columns, *baseline_reading, post_reading, post_math, student_id, school_id, grade_level,* and *passed* were removed as they were either outside the scope of this study or posed a risk of data leakage. The *post_math* column was specifically excluded to prevent data leakage, as it directly reflects the outcome being measured. The cleaned dataset was verified using *dim(), colnames(),* and *head()* functions, confirming a final dataset of 10,000 rows and 10 columns with all variables in the correct numerical format, ready for regression analysis (Wickham, H. 2016).

```
### Remove Irrelevant Columns
```{r remove_columns}
student_math <- student %>%
  select(-baseline_reading, -post_reading, -post_math,
         -student_id, -school_id, -grade_level, -passed)
```
```

```
[1] 10000    10
 [1] "intervention"            "ses_index"               "attendance_rate"
"study_hours_wk"
 [5] "teacher_experience_years" "class_size"              "baseline_math"
"baseline_motivation_1_10"
 [9] "test_anxiety_1_10"       "math_improvement"
```

	intervention <dbl>	ses_index <dbl>	attendance_rate <dbl>	study_hours_wk <dbl>	teacher_experience_years <dbl>
1	1	1.379	0.996	5.93	0.0
2	1	2.037	0.930	2.52	6.7
3	4	0.403	0.933	5.32	8.3
4	3	-0.442	0.924	5.26	9.2
5	4	-0.572	0.971	7.47	10.6
6	1	-1.128	0.875	4.12	15.7

6 rows | 1-6 of 10 columns

class_size <int>	baseline_math <dbl>	baseline_motivation_1_10 <dbl>	test_anxiety_1_10 <dbl>	math_improvement <dbl>
28	76.9	7.7	2.6	10.9
32	78.9	7.1	4.3	10.0
23	69.4	5.8	3.8	15.4
30	55.9	6.7	5.6	16.8
21	55.3	4.6	7.6	20.4
25	59.4	7.4	6.8	9.7

6 rows | 7-11 of 10 columns

D. DATA ANALYSIS

This study employed multiple linear regression as the primary analytical technique, supplemented by stepwise regression for variable selection, a Q-Q plot, a histogram, and four residual diagnostic plots for normality and assumption assessment, a VIF test for multicollinearity detection, 10-fold cross-validation, an 80/20 train-test split for model stability assessment, Cook's Distance, and a box plot for outlier detection. All analyses were performed in RStudio using base R functions and the *tidyverse, ggplot2, car,* and *caret* packages (Wickham, H. 2016).

Prior to building the regression model, exploratory data analysis was conducted to understand the distribution of the dependent variable and visualize the relationship between intervention type and math score improvement.

Summary Statistics

Summary statistics of the cleaned dataset showed an average improvement in math scores of 10.87 points, with a median of 11.00, indicating a symmetric distribution. The average student attendance rate was 91.3%, with students studying an average of 4.58 hours per week. The average teaching experience was 9.06 years, and the average class size was 26 students. Baseline math scores averaged 70.03 out of 100, suggesting a moderate starting point across the student population.

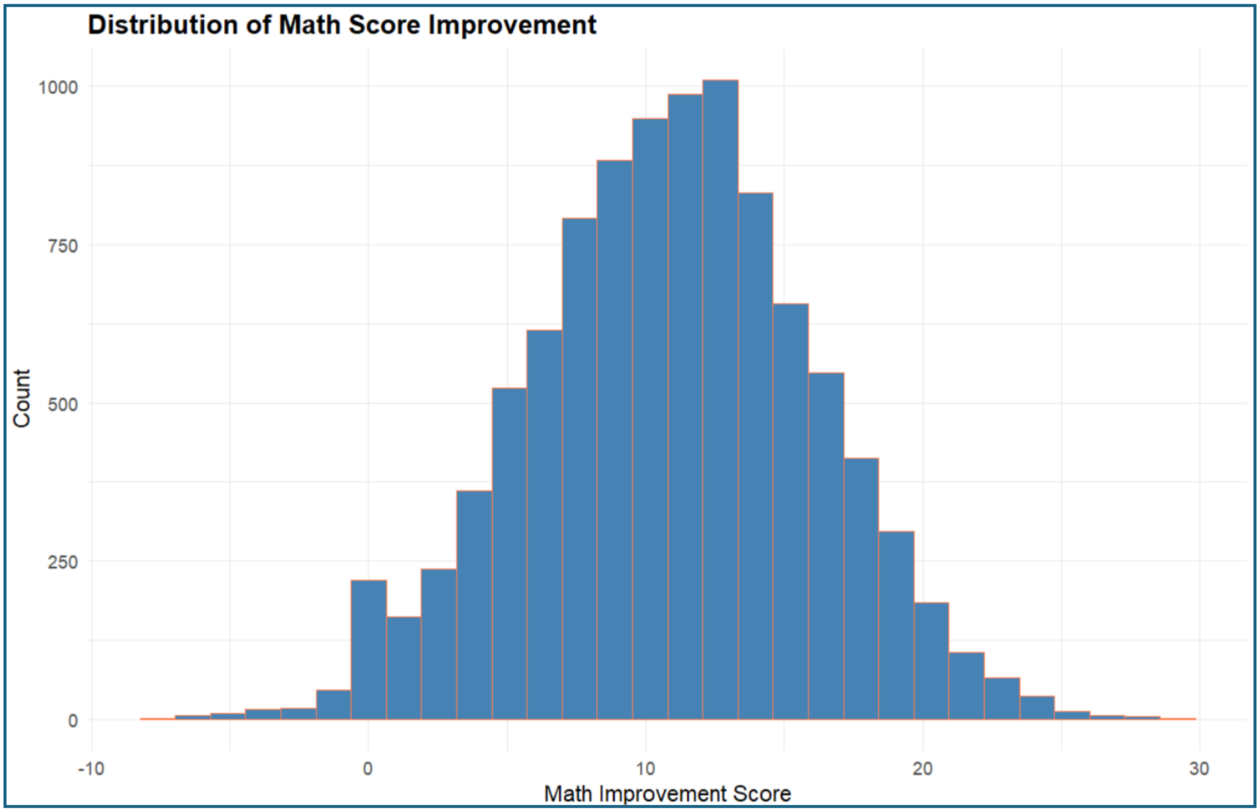
```
## DATA ANALYSIS
### Summary of the Data
```{r summary_data}
summary(student_math)
```
```

```
  intervention     ses_index          attendance_rate  study_hours_wk   teacher_experience_years
 class_size
  Min.   :1.000   Min.   :-2.500000   Min.   :0.6670   Min.   : 0.020   Min.   : 0.000           Min.
 :10.00
  1st Qu.:2.000   1st Qu.:-0.678250   1st Qu.:0.8730   1st Qu.: 2.250   1st Qu.: 5.600           1st
 Qu.:22.00
  Median :2.000   Median : 0.001000   Median :0.9160   Median : 3.910   Median : 9.000           Median
 :26.00
  Mean   :2.512   Mean   :-0.002791   Mean   :0.9129   Mean   : 4.576   Mean   : 9.057           Mean
 :26.06
  3rd Qu.:4.000   3rd Qu.: 0.671000   3rd Qu.:0.9570   3rd Qu.: 6.140   3rd Qu.:12.400           3rd
 Qu.:30.00
  Max.   :4.000   Max.   : 2.500000   Max.   :1.0000   Max.   :22.850   Max.   :28.700           Max.
 :40.00
  baseline_math    baseline_motivation_1_10 test_anxiety_1_10 math_improvement
  Min.   : 20.00   Min.   : 1.000           Min.   : 1.000    Min.   :-7.20
  1st Qu.: 61.00   1st Qu.: 5.300           1st Qu.: 3.700    1st Qu.: 7.40
  Median : 70.00   Median : 6.500           Median : 4.800    Median :11.00
  Mean   : 70.03   Mean   : 6.456           Mean   : 4.764    Mean   :10.87
  3rd Qu.: 79.20   3rd Qu.: 7.600           3rd Qu.: 5.900    3rd Qu.:14.30
  Max.   :100.00   Max.   :10.000           Max.   :10.000    Max.   :29.60
```

Histogram: Distribution of Math Score Improvement

A histogram was generated to visualize the distribution of math score improvement across all 10,000 student records. The histogram revealed that math score improvement follows approximately a normal distribution, with a mean improvement of 10.87 points and a range of -7.20 to 29.60 points, with most students improving by 8 to 15 points. The approximately normal distribution of the dependent variable supports the use of multiple linear regression and provides initial confirmation that the normality assumption required for regression analysis was met.

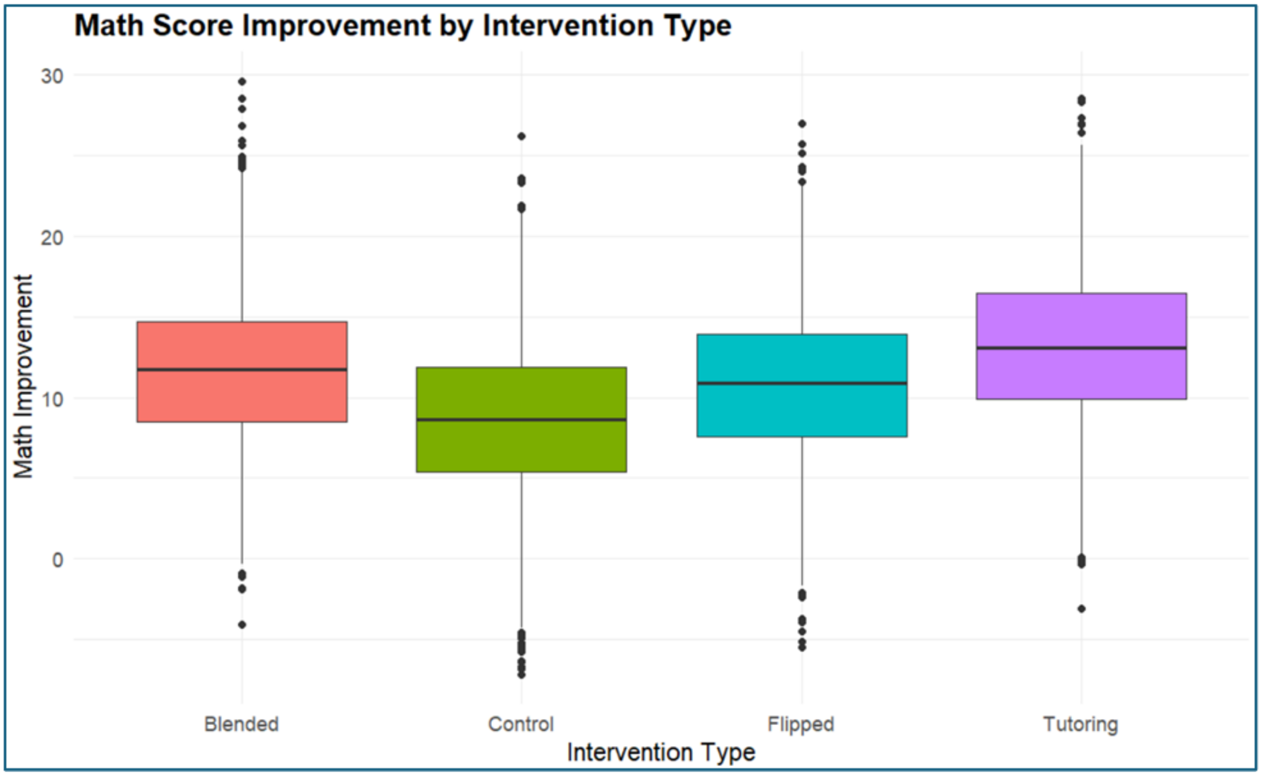
```
## PLOTTING THE DATA
### Distribution of Math Improvement
```{r eda_histogram, fig.width=10, fig.height=6, fig.align='center'}
ggplot(student_math, aes(x = math_improvement)) +
  geom_histogram(fill = "steelblue", color = "coral", bins = 30) +
  labs(title = "Distribution of Math Score Improvement",
       x = "Math Improvement Score",
       y = "Count") +
  theme_minimal() +
  theme(plot.title = element_text(size = 14, face = "bold"),
        axis.title = element_text(size = 12),
        axis.text = element_text(size = 10))
```
```



Boxplot: Math Score Improvement by Intervention Type

A boxplot was generated to visually compare the improvement in math scores across the four intervention types prior to regression analysis. The results revealed that tutoring produced the largest median improvement in math scores at approximately 13 points, followed by blended learning at approximately 12 points, the flipped classroom at approximately 11 points, and the control group at approximately 9 points. All three active intervention methods outperformed the control group, which received traditional instruction with no specialized intervention. Additionally, all four groups contained outliers both above and below the interquartile range, indicating natural variability in student performance regardless of intervention type. Consistent with standard practice in large-sample regression analysis, outliers were retained in the dataset as they reflect natural variability in student performance, and their removal could introduce bias into the findings.

```
### Math Improvement by 'Intervention Type'
```{r eda_boxplot, fig.width=10, fig.height=6, fig.align='center'}
ggplot(student, aes(x = factor(intervention), y = math_improvement,
                    fill = factor(intervention))) +
  geom_boxplot() +
  scale_x_discrete(labels = c("1" = "Blended", "2" = "Control",
                              "3" = "Flipped", "4" = "Tutoring")) +
  labs(title = "Math Score Improvement by Intervention Type",
       x = "Intervention Type",
       y = "Math Improvement",
       fill = "Intervention") +
  theme_minimal() +
  theme(plot.title = element_text(size = 14, face = "bold"),
        axis.title = element_text(size = 12),
        axis.text = element_text(size = 10),
        legend.position = "none")
```
```



Full Regression Model

A full multiple linear regression model was constructed using all nine predictor variables to predict math score improvement. Multiple linear regression was selected because it simultaneously evaluates the relationships among multiple independent variables and a single continuous dependent variable, producing interpretable coefficients that quantify each predictor's contribution. It is the most appropriate technique for this study, given the continuous nature of the dependent variable and the presence of multiple continuous predictor variables. The full model produced an R-Squared of 0.1376 (13.76%), an Adjusted R-Squared of 0.1368, and an F-statistic of 177 ($p < 2.2e-16$).

(Thulin, M. 2026).

```
## REGRESSION MODEL
### Build the Model
```{r full_model}
model_full <- lm(math_improvement ~ ., data = student_math)
summary(model_full)
```
```

```
Call:
lm(formula = math_improvement ~ ., data = student_math)

Residuals:
     Min       1Q   Median       3Q      Max 
-18.6904  -3.2260   0.0832   3.2278  19.3880 

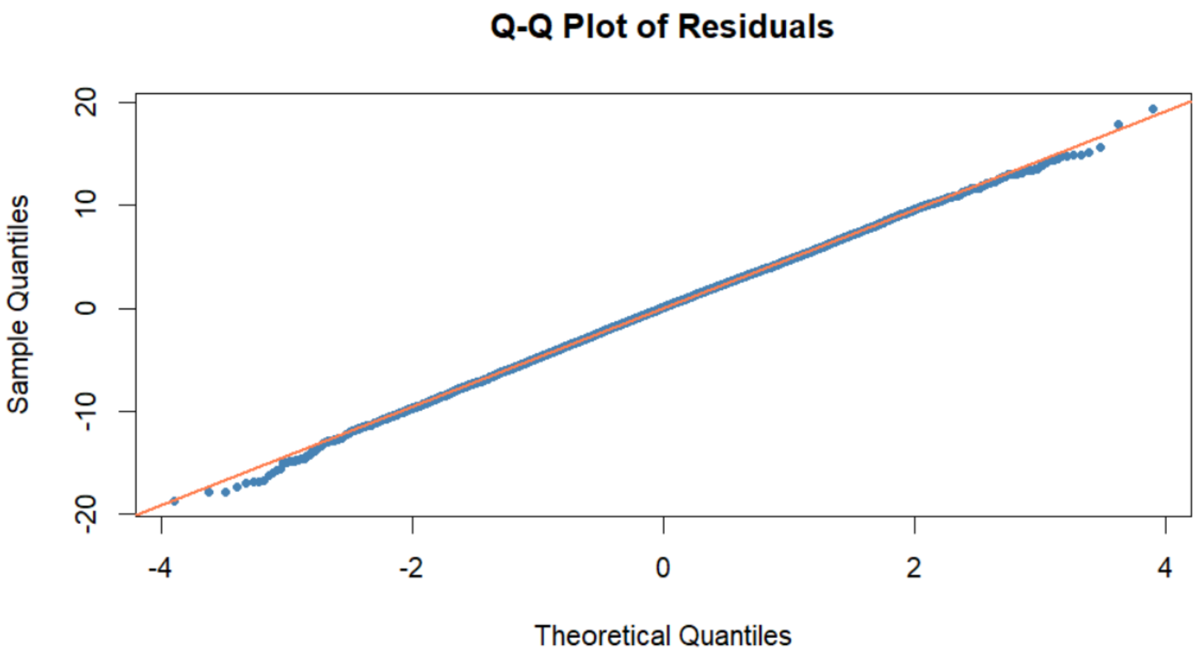
Coefficients:
                          Estimate Std. Error t value Pr(>|t|)    
(Intercept)              11.660815   0.905258  12.881  < 2e-16 ***
intervention              0.707824   0.043671  16.208  < 2e-16 ***
ses_index                 0.030308   0.060726   0.499    0.618    
attendance_rate           5.267672   0.845315   6.232  4.8e-10 ***
study_hours_wk            0.307931   0.015935  19.324  < 2e-16 ***
teacher_experience_years  0.121575   0.009955  12.213  < 2e-16 ***
class_size               -0.117643   0.008055 -14.604  < 2e-16 ***
baseline_math            -0.091018   0.004017 -22.660  < 2e-16 ***
baseline_motivation_1_10 -0.064838   0.033223  -1.952    0.051 .  
test_anxiety_1_10        -0.005873   0.030449  -0.193    0.847    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 4.801 on 9990 degrees of freedom
Multiple R-squared:  0.1376,	Adjusted R-squared:  0.1368 
F-statistic:   177 on 9 and 9990 DF,  p-value: < 2.2e-16
```

Q-Q Plot of Residuals

The Q-Q plot showed that the residuals closely followed the normal distribution line throughout the central region, with only minor deviations at the end. This confirmed that the normality assumption for multiple linear regression was sufficiently met (Thulin, M. 2026).

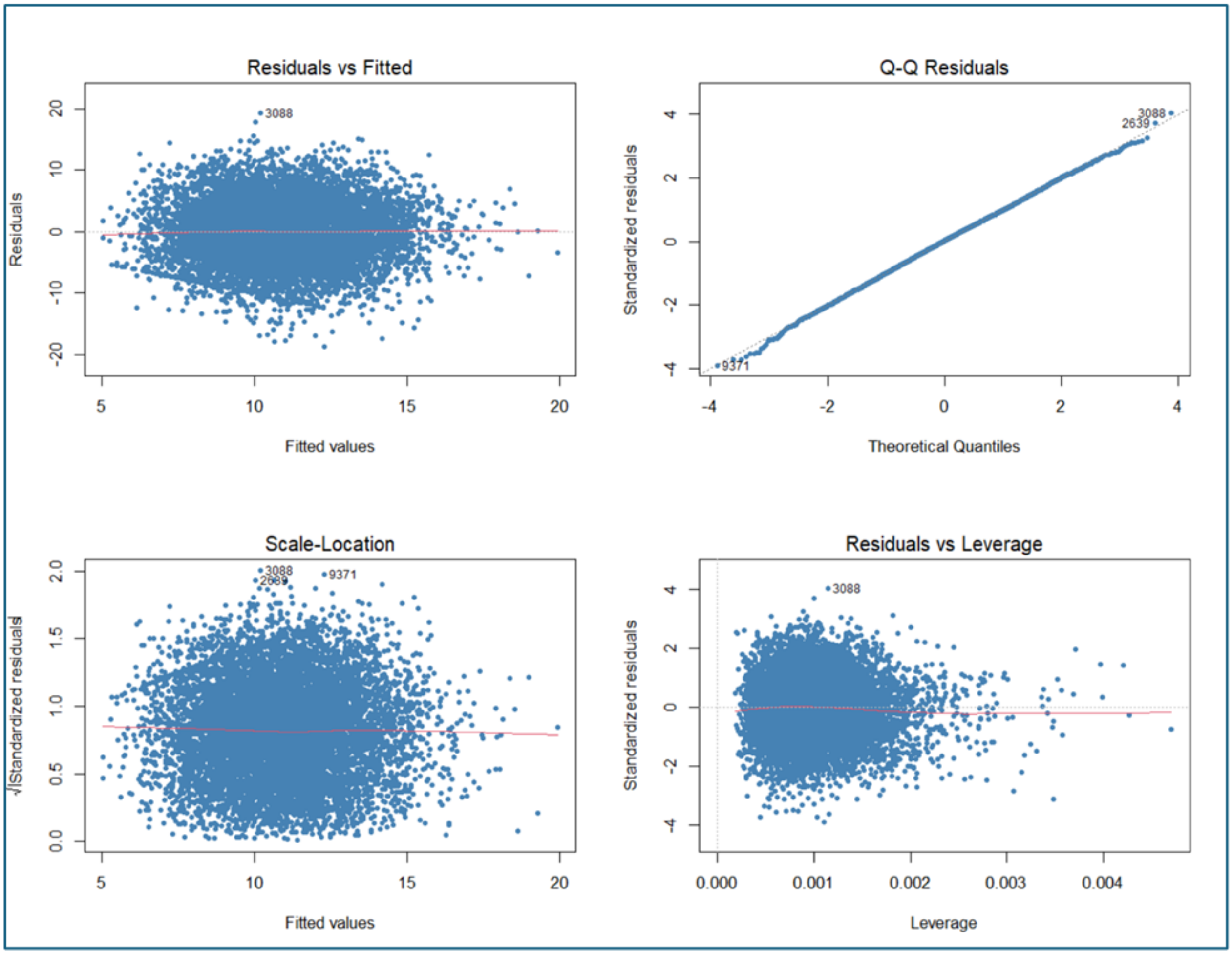
```
### Q-Q Plot of Residuals
```{r qq_plot, fig.width=10, fig.height=6, fig.align='center'}
qqnorm(residuals(model_full),
       main = "Q-Q Plot of Residuals",
       xlab = "Theoretical Quantiles",
       ylab = "Sample Quantiles",
       col  = "steelblue",
       pch  = 20)
qqline(residuals(model_full), col = "coral", lwd = 2)
```
```



Residual Diagnostic Plots

Four residual diagnostic plots were examined to validate the regression assumptions. Results confirmed that the linearity, normality, and homoscedasticity assumptions were met, and no single observation had excessive influence on the model (Thulin, M. 2026).

```
### Residual Diagnostic Plots
```{r residual_plots, fig.width=10, fig.height=8, fig.align='center'}
par(mfrow = c(2, 2))
plot(model_full, col = "steelblue", pch = 20)
par(mfrow = c(1, 1))
```
```



VIF Multicollinearity Test

A Variance Inflation Factor (VIF) test was conducted to detect multicollinearity among the predictor variables. All VIF values ranged from 1.001 to 1.562, well below the threshold of 5, confirming that none of the predictor variables were redundant or highly correlated with one another. Each variable contributes unique and independent information to the model (Thulin, M. 2026).

```
### VIF Multicollinearity Test
```{r vif_test}
vif(model_full)
```

            intervention   ses_index   attendance_rate   study_hours_wk
                1.001329   1.562220   1.010798   1.083465
 teacher_experience_years   class_size   baseline_math baseline_motivation_1_10
                1.000554   1.000807   1.235622   1.288796
         test_anxiety_1_10
                1.060669
```

Stepwise Regression

Bidirectional stepwise regression was applied using the AIC criterion to identify the most statistically significant predictors and produce a final model. Stepwise regression automatically removed two non-significant predictors, *ses_index* (AIC = 31381) and *test_anxiety_1_10* (AIC = 31383), reducing the model from nine to seven predictors while improving the F-statistic from 177 to 227.6. The final stepwise model produced an R-Squared of 0.1375 (13.75%) and an Adjusted R-Squared of 0.1369 (Thulin, M. 2026).

````
### Stepwise Regression Model
```{r}
model_stepwise <- step(model_full, direction = "both")
summary(model_stepwise)
```
````

```
Start:  AIC=31385.13
math_improvement ~ intervention + ses_index + attendance_rate +
    study_hours_wk + teacher_experience_years + class_size +
    baseline_math + baseline_motivation_1_10 + test_anxiety_1_10

                           Df Sum of Sq    RSS   AIC
- test_anxiety_1_10         1       0.9 230235 31383
- ses_index                 1       5.7 230240 31383
<none>                                  230234 31385
- baseline_motivation_1_10  1      87.8 230322 31387
- attendance_rate           1     895.0 231129 31422
- teacher_experience_years  1    3437.4 233672 31531
- class_size                1    4915.6 235150 31594
- intervention              1    6054.5 236289 31643
- study_hours_wk            1    8606.0 238840 31750
- baseline_math             1   11834.0 242069 31884
```

```
Step:  AIC=31383.17
math_improvement ~ intervention + ses_index + attendance_rate +
    study_hours_wk + teacher_experience_years + class_size +
    baseline_math + baseline_motivation_1_10

                           Df Sum of Sq    RSS   AIC
- ses_index                 1       6.4 230242 31381
<none>                                  230235 31383
- baseline_motivation_1_10  1      87.0 230322 31385
+ test_anxiety_1_10         1       0.9 230234 31385
- attendance_rate           1     895.5 231131 31420
- teacher_experience_years  1    3436.6 233672 31529
- class_size                1    4918.7 235154 31593
- intervention              1    6055.9 236291 31641
- study_hours_wk            1    8605.2 238841 31748
- baseline_math             1   11839.9 242075 31883
```

```
Step:  AIC=31381.45
math_improvement ~ intervention + attendance_rate + study_hours_wk + 
    teacher_experience_years + class_size + baseline_math + baseline_motivation_1_10

                           Df Sum of Sq    RSS   AIC
<none>                                  230242 31381
- baseline_motivation_1_10  1      81.8 230324 31383
+ ses_index                 1       6.4 230235 31383
+ test_anxiety_1_10         1       1.5 230240 31383
- attendance_rate           1     914.8 231157 31419
- teacher_experience_years  1    3438.1 233680 31528
- class_size                1    4919.3 235161 31591
- intervention              1    6058.1 236300 31639
- study_hours_wk            1    8908.0 239150 31759
- baseline_math             1   13782.1 244024 31961
```

```
Call:
lm(formula = math_improvement ~ intervention + attendance_rate + 
    study_hours_wk + teacher_experience_years + class_size + 
    baseline_math + baseline_motivation_1_10, data = student_math)

Residuals:
     Min       1Q   Median       3Q      Max 
-18.6380  -3.2214   0.0865   3.2343  19.3840 

Coefficients:
                          Estimate Std. Error t value Pr(>|t|)    
(Intercept)              11.485719   0.847251  13.556  < 2e-16 ***
intervention              0.708008   0.043665  16.214  < 2e-16 ***
attendance_rate           5.306551   0.842221   6.301 3.09e-10 ***
study_hours_wk            0.309238   0.015728  19.662  < 2e-16 ***
teacher_experience_years  0.121573   0.009953  12.215  < 2e-16 ***
class_size               -0.117673   0.008054 -14.611  < 2e-16 ***
baseline_math            -0.090191   0.003688 -24.456  < 2e-16 ***
baseline_motivation_1_10 -0.057404   0.030468  -1.884   0.0596 .  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 4.8 on 9992 degrees of freedom
Multiple R-squared:  0.1375,	Adjusted R-squared:  0.1369 
F-statistic: 227.6 on 7 and 9992 DF,  p-value: < 2.2e-16
```

Cross Validation

Ten-fold cross-validation was conducted to assess the model's generalizability to unseen data. The cross-validated R-Squared of 13.72% was highly consistent with the stepwise model R-Squared of 13.75%, confirming that the model is stable and does not overfit.

```
### Cross Validation
```{r cross_validation}
set.seed(123)
train_control <- trainControl(method = "cv", number = 10)
cv_model <- train(math_improvement ~ intervention +
                  attendance_rate + study_hours_wk +
                  teacher_experience_years + class_size +
                  baseline_math + baseline_motivation_1_10,
                  data = student_math,
                  method = "lm",
                  trControl = train_control)
print(cv_model)
cat("Cross Validated R-Squared:",
    round(cv_model$results$Rsquared, 4), "\n")
cat("Cross Validated R-Squared %:",
    round(cv_model$results$Rsquared * 100, 2), "%\n")
```
```

```
Linear Regression

10000 samples
    7 predictor

No pre-processing
Resampling: Cross-Validated (10 fold)
Summary of sample sizes: 9000, 9000, 8998, 9001, 9000, 9000, ...
Resampling results:

  RMSE      Rsquared  MAE
  4.801165  0.1372    3.827075

Tuning parameter 'intercept' was held constant at a value of TRUE
Cross Validated R-Squared: 0.1372
Cross Validated R-Squared %: 13.72 %
```

Train-Test Split

An 80/20 train-test split was conducted to further evaluate model performance on unseen data. The model trained on 80% of the data (8,001 records) achieved a training R-Squared of 13.84%, while the test R-Squared on the remaining 20% (1,999 records) was 13.33%. The minimal difference of 0.51% between the training and test R-Squared values confirms that the model generalizes well with new data (Thulin, M. 2026).

.

```
### Train Test Split
```{r train_test_split}
set.seed(123)
train_index <- createDataPartition(student_math$math_improvement,
                                   p = 0.80, list = FALSE)
train_data <- student_math[train_index, ]
test_data  <- student_math[-train_index, ]

model_train <- lm(math_improvement ~ intervention +
                  attendance_rate + study_hours_wk +
                  teacher_experience_years + class_size +
                  baseline_math + baseline_motivation_1_10,
                  data = train_data)

predictions <- predict(model_train, newdata = test_data)

ss_res <- sum((test_data$math_improvement - predictions)^2)
ss_tot <- sum((test_data$math_improvement -
               mean(test_data$math_improvement))^2)
r2_test <- 1 - (ss_res / ss_tot)

cat("Training R-Squared:", round(summary(model_train)$r.squared, 4), "\n")
cat("Test R-Squared:     ", round(r2_test, 4), "\n")
cat("Test R-Squared %:   ", round(r2_test * 100, 2), "%\n")
```

 Training R-Squared: 0.1384
 Test R-Squared: 0.1333
 Test R-Squared %:   13.33 %
```

Outlier Detection

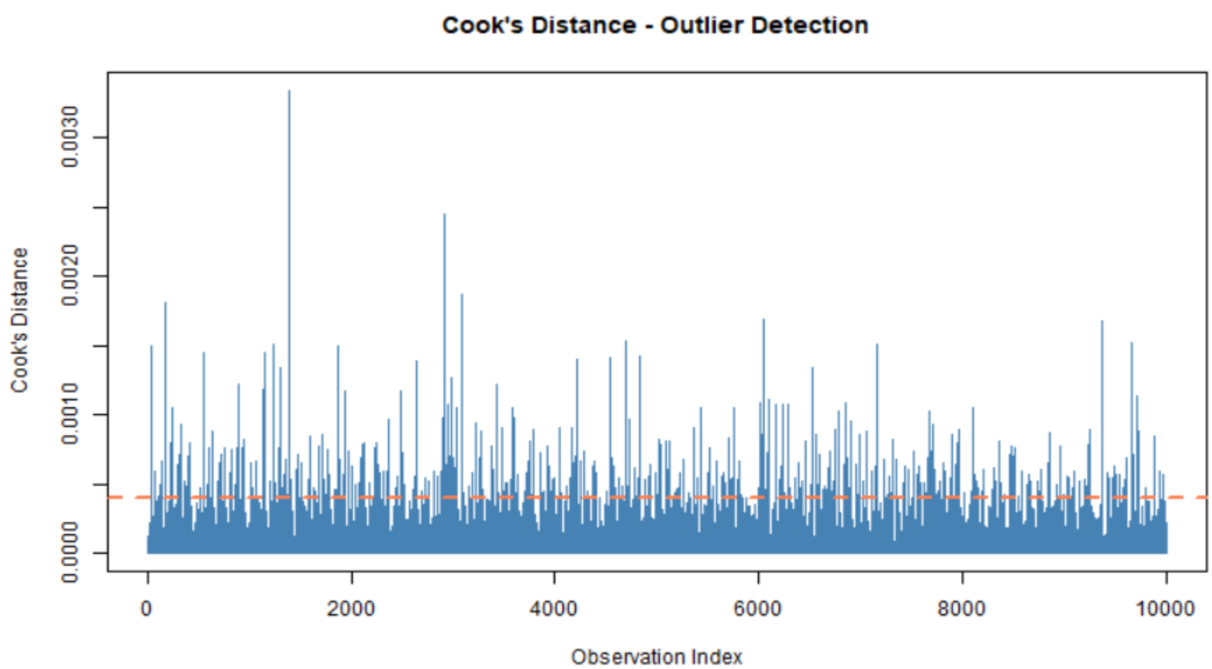
Cook's Distance was calculated to identify influential observations. While all observations' Cook's Distance values remained below the conservative threshold of 1.0, 517 observations (5.17%) were flagged as potentially influential using the standard threshold of 4/n. These observations were retained to avoid introducing selection bias, as in large samples of 10,000 records, observations exceeding the 4/n threshold are not necessarily problematic. However, their presence warrants consideration when interpreting the model's results, as they may contribute to slight sensitivity in the regression coefficients.

~~~~
### Outlier Detection
```{r outlier_check, fig.width=10, fig.height=6, fig.align='center'}
cooks_d <- cooks.distance(model_full)
influential <- which(cooks_d > 4/nrow(student_math))
cat("Number of influential outliers:", length(influential), "\n")
cat("Percentage of dataset:          ",
    round(length(influential)/nrow(student_math)*100, 2), "%\n")

plot(cooks_d, type = "h",
     main = "Cook's Distance - Outlier Detection",
     xlab = "Observation Index",
     ylab = "Cook's Distance",
     col = "steelblue")
abline(h = 4/nrow(student_math), col = "coral", lwd = 2, lty = 2)
```
~~~~

R Console

```
Number of influential outliers: 517
Percentage of dataset:          5.17 %
```




~~~~
## MODEL EVALUATION
### Extract R-Squared Values
```{r r_squared}
r2_full <- summary(model_full)$r.squared
r2_stepwise <- summary(model_stepwise)$r.squared
```
~~~~

```
### Adjusted R-Squared
```{r adusted_rsquared}
adj_r2_full <- summary(model_full)$adj.r.squared
adj_r2_stepwise <- summary(model_stepwise)$adj.r.squared
```
```

```
### Print Results
```{r print_result}
cat("=== Full Model ===\n")
cat("R-Squared:          ", round(r2_full, 4), "\n")
cat("R-Squared %:        ", round(r2_full * 100, 2), "%\n")
cat("Adjusted R-Squared:", round(adj_r2_full, 4), "\n")

cat("\n=== Stepwise Model ===\n")
cat("R-Squared:          ", round(r2_stepwise, 4), "\n")
cat("R-Squared %:        ", round(r2_stepwise * 100, 2), "%\n")
cat("Adjusted R-Squared:", round(adj_r2_stepwise, 4), "\n")

cat("\n=== Hypothesis Decision ===\n")
if(r2_stepwise > 0.70) {
  cat("R-Squared > 0.70: REJECT H0 - Model exceeds 70% accuracy threshold\n")
} else {
  cat("R-Squared <= 0.70: FAIL TO REJECT H0 - Model does not exceed 70% threshold\n")
}
```
```

```
=== Full Model ===
R-Squared:          0.1376
R-Squared %:        13.76 %
Adjusted R-Squared: 0.1368

=== Stepwise Model ===
R-Squared:          0.1375
R-Squared %:        13.75 %
Adjusted R-Squared: 0.1369

=== Hypothesis Decision ===
R-Squared <= 0.70: FAIL TO REJECT H0 - Model does not exceed 70% threshold
```

Advantages of Analytical Technique

A significant advantage of multiple linear regression combined with stepwise selection is its interpretability. Multiple linear regression produces explicit coefficients for each predictor variable that clearly indicate the direction and magnitude of each variable's effect on the dependent variable. This transparency is particularly valuable in an educational research context, where administrators and educators need to understand not only what the model predicts but also why, enabling informed, evidence-based decision-making regarding instructional strategies and resource allocation (Aissaoui et al., 2020).

Disadvantage of Analytical Technique

A disadvantage of multiple linear regression is its assumption of linearity. The technique assumes a straight-line relationship between each predictor and the dependent variable. If the true relationship between predictors and math score improvement is non-linear, the model will fail to capture these patterns, resulting in reduced predictive accuracy. The low R-Squared of 13.75%

observed in this study suggests that linear regression may not fully capture the complexity of the relationships in the data, and that non-linear modeling techniques such as Random Forest or Gradient Boosting may be more appropriate for future analyses of this dataset.

E. DATA SUMMARY AND IMPLICATIONS

This study investigated whether a multiple linear regression model can be constructed on a student success dataset to statistically quantify the impact of academic intervention methods on student math score improvement. The analysis confirmed that a statistically significant multiple linear regression model can be constructed, with the final stepwise regression model producing an F-statistic of 227.6 ($p < 2.2e-16$), indicating the model is highly significant overall. However, the model's R-Squared of 13.75% did not meet the 70% predictive accuracy threshold specified in the alternate hypothesis, leading to a failure to reject the null hypothesis.

Despite not meeting the threshold, the analysis yielded interpretable findings directly relevant to the research question. Six statistically significant predictors of math score improvement were identified: *intervention type* (Beta = 0.708, $p < 0.001$), *attendance rate* (Beta = 5.307, $p < 0.001$), *weekly study hours* (Beta = 0.309, $p < 0.001$*), teacher experience years* (Beta = 0.122, $p < 0.001$), *class size* (Beta = -0.118, $p < 0.001$), and *baseline math score* (Beta = -0.090, $p < 0.001$). Exploratory data analysis further revealed that *tutoring* produced the largest median improvement in math scores, followed by *blended* learning, the *flipped classroom*, and the *control* group receiving traditional instruction. These findings directly address the research question by confirming that intervention type is a statistically significant predictor of student math score improvement and that *tutoring* is the most effective intervention method among those examined.

The consistency of R-Squared values across all evaluation methods, full model (13.76%), stepwise model (13.75%), cross-validation (13.72%), and train-test split test data (13.33%), confirms that the model is stable, does not overfit, and generalizes reliably to new data. Cook's Distance analysis identified 517 influential observations (5.17%), all of which were retained, as none exceeded the conservative threshold of 1.0, indicating no extreme leverage points in the dataset.

A limitation of the analysis is the low R-Squared of 13.75%, which indicates that the identified predictors account for only a small portion of the total variance in student math score improvement. This suggests that a substantial proportion of the variance is attributable to unmeasured factors not captured in the dataset, such as parental involvement, student mental health, learning disabilities, and home environment. Furthermore, the intervention groups may not have been randomly assigned, which introduces the possibility of selection bias, as pre-existing differences between student groups may have influenced the observed outcomes independently of the intervention method.

Based on this analysis, schools and educational administrators should prioritize tutoring as the primary academic intervention for students seeking to improve their math scores. It is important to note that the observed effectiveness of tutoring may be influenced by factors beyond the tutoring method itself. Students participating in tutoring programs may also receive additional support at home, demonstrate higher levels of motivation, or have self-selected into tutoring due to pre-existing academic engagement. These confounding factors may have contributed to the observed improvement and should be considered when interpreting the tutoring findings.

In addition to selecting the appropriate intervention method, administrators should focus on the following evidence-based strategies to maximize student math performance by improving attendance policies, as attendance rate was the strongest predictor of math improvement (Beta = 5.307), meaning even small improvements in attendance could yield significant academic gains; encouraging structured study time, as students who studied more hours per week showed greater math improvement; reducing class sizes, as larger class sizes were negatively associated with math improvement; and prioritizing experienced teachers, as teacher experience was positively associated with math improvement. These recommendations are grounded in the regression model's statistical evidence and provide schools with actionable, data-driven guidance for instructional planning and resource allocation (Aissaoui et al., 2020).

Future research should incorporate additional variables such as parental involvement, learning disabilities, and socioeconomic conditions to improve model accuracy and generalizability. Additionally, applying non-linear modeling techniques such as Random Forest or Gradient Boosting to the same dataset may capture complex relationships not detectable by linear regression and yield significantly higher predictive accuracy.

F. SOURCES

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